European Journal of Operational Research 326 (2025) 255–269

Contents lists available at ScienceDirect

European Journal of Operational Research

journal homepage: www.elsevier.com/locate/ejor

Production, Manufacturing, Transportation and Logistics

Capacitated hub location routing problem with time windows and stochastic demands for the design of intra-city express systems

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ARTICLE INFO

Keywords:
Intra-city express
Multi-stage recourse model
Hybrid stochastic variable neighbourhood search algorithm
Case studies
Stochastic programming

ABSTRACT

This work focuses on planning an intra-city express system in a practical environment. Various operation characteristics, such as vehicle capacity, hub capacity, time windows, and stochastic demands, have been considered. Therefore, we introduce a capacitated hub location routing problem with time windows and stochastic demand and formulate it using a multi-stage recourse model. In this model, long-term decisions (hub location and client-to-hub allocation) are made first, and short-term decisions (vehicle routing) are determined after revealing stochastic variables. To solve the problem, we propose a hybrid stochastic variable neighbourhood search (HSVNS) algorithm, which integrates an adaptive large neighbourhood search (ALNS) algorithm within a stochastic variable neighbourhood search (SVNS) framework. Numerical experiments and case studies indicate that the HSVNS algorithm can provide high-quality solutions within a reasonable computation time for instances with up to 70 clients and that considering stochastic factors can efficiently reduce operation costs, especially for instances with tight vehicle capacity and loose time windows.

1. Introduction

With the rise of e-commerce, intra-city express services have become an essential part of urban logistics systems. Various express companies are offering services like “delivery within 24 h”, “same day delivery within the city”, or “next day delivery”, e.g., SF Express, Yamato Transport, Japan Post, and so on. Therefore, providing intra-city express services cost-effectively has become a significant challenge for these companies. This issue is also a growing concern for urban governments, as intra-city express deliveries have led to problems such as traffic congestion and air pollution (Zhao et al., 2019; Yıldız & Savelsbergh, 2022).

This study concentrates on the design of an intra-city express system within a practical environment. The target system, illustrated in Fig. 1, is structured as a less-than-truckload hub-and-spoke network in which local tours connect hubs and branch offices instead of direct links (Kartal et al., 2017; Alumur et al., 2021). The parcels are collected at the origin branch offices, sorted at the first hub, possibly transported to the second hub, and delivered to the destination branch offices. In addition, the pickup and delivery processes are conducted simultaneously along local tours.

Branch offices often lack sufficient sorting resources (e.g., labour, machinery, space, and so on). Consequently, parcels are collected in a mixed state and must be sorted by destination at hubs for further delivery, i.e., the flow cannot be exchanged directly, even within the same local tour. More specifically, in the current service cycle (e.g., this morning), each vehicle departs from its hub and traverses a subset of branch offices, distributing parcels collected in the previous service cycle (e.g., yesterday morning) and collecting parcels to be delivered in the upcoming service cycle (e.g., tomorrow morning). Thus, the service system operates as a warmed-up transportation system. Furthermore, backbone (inter-hub) transportation occurs after the vehicles return to the hubs (e.g., at night). A detailed illustration of this process is provided in Fig. 2. Similar assumptions have been adopted in various studies on the intra-city express system design (such as Sun, 2013; Karimi, 2018; Wu et al., 2024).

Based on the above descriptions, one can find that the key decisions involved in designing the target system include hub location, branch-office-to-hub allocation, and vehicle routing, which should be addressed jointly. Moreover, the following four practical conditions are

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https://doi.org/10.1016/j.ejor.2025.05.006
Received 1 September 2024; Accepted 2 May 2025
Available online 5 May 2025
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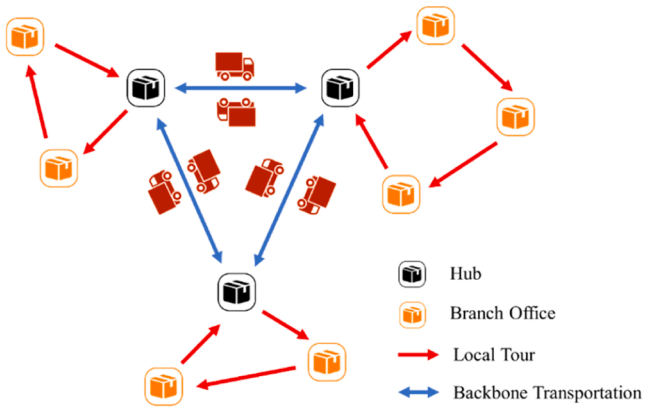


Fig. 1. Hub-and-spoke network for intra-city express systems.

taken into account:

i) Capacity. Both hub and vehicle capacities must be considered due to the limited land resources and the restriction on using large-volume trucks in urban areas.
ii) Time window. Intra-city express service users often have strict requirements on delivery times, making it necessary to impose time window constraints.
iii) Single-allocation. In practical applications, each branch office is typically assigned to a single hub, as branch offices generally lack sufficient sorting capacities.
iv) Stochastic demand. The express company often lacks prior knowledge of parcel flows since business activities might introduce uncertainty into these flows, i.e., the intra-city express demands are stochastic rather than deterministic. One natural approach to manage this uncertainty is to make hub location and branch-office-to-hub allocation decisions (long-term decisions) before any random variables are revealed (i.e., before the start of service cycles), as altering these decisions for an established system is costly. In each service cycle, vehicle routing (short-term decisions) is determined based on the known distribution demand (since these parcels have been collected in the previous service cycle) and the unknown collection demand. Finally, recourse actions and inter-hub transportation are implemented to complete the distribution process, as described in Fig. 3.

With these considerations, we propose the capacitated hub location routing problem with time windows and stochastic demand (CHLRPTWSD) to design intra-city express systems. This problem falls within the single-allocation hub location routing problem (SAHLRP, Catanzaro et al., 2011; Fontes & Goncalves, 2021), which will be thoroughly reviewed in Section 2.

The key contributions of this study are fourfold: i) A multi-stage recourse model is introduced to formulate the CHLRPTWSD, which models the stochastic SAHLRP via the recourse model for the first time. ii) An HSVNS algorithm, which integrates an ALNS algorithm within an SVNS framework, is proposed to solve the problem. iii) Numerical experiments are conducted on modified benchmark instances to demonstrate the efficiency of the proposed framework. iv) Finally, case studies are generated using real-world data, and the results are presented to highlight the practical applicability and usefulness of the proposed model and algorithm.

The remainder of the paper is structured as follows: Section 2 reviews the SAHLRP and compares our study with the existing literature. Section 3 defines the CHLRPTWSD via a multi-stage recourse model. Section 4 introduces the HSVNS algorithm, whose efficiency is proved in Section 5. The results of the case studies are presented in Section 6, followed by Section 7 as a conclusion of the study.

2. Literature review

This section mainly reviews studies on the SAHLRP, which is highly

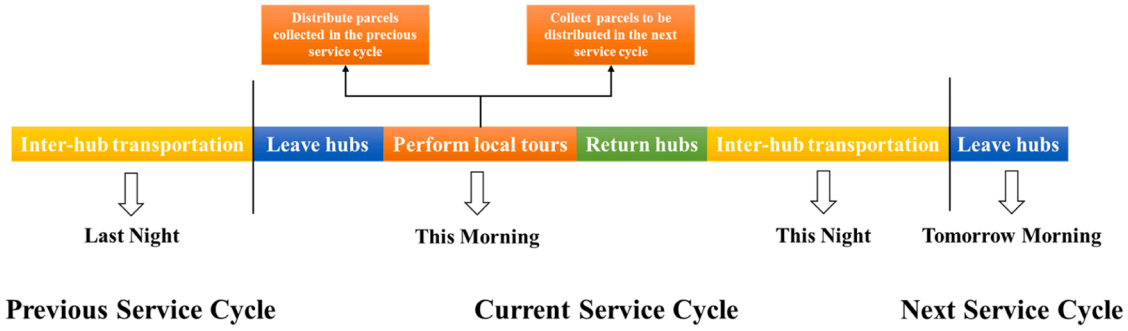


Fig. 2. Service procedures for intra-city express systems.

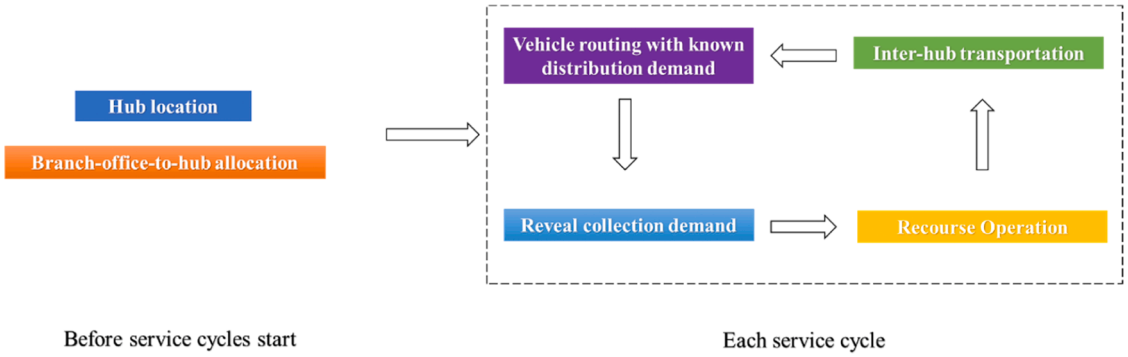


Fig. 3. Stochastic decision process for intra-city express systems.

relevant to the CHLRPTWSD. Nagy and Salhi (1998) first proposed the SAHLRP with route length constraints to limit working hours. They developed a non-polynomial-sized linear programming model and tackled the problem via a two-stage algorithm on an instance with 249 clients.

So far, most research on the SAHLRP has concentrated on postal service networks, where the collection and distribution processes usually co-occur. Bostel et al. (2015) examined an SAHLRP, where the maximum number of visited clients constrained the length of each vehicle route. They introduced a memetic algorithm (MA) to solve instances with up to 100 clients and compared its efficiency with the CPLEX solver. Kartal et al. (2017) investigated the operational characteristics of a major cargo company in Turkey, proposing three variants of single-allocation p-hub median location routing formulations. They employed a multi-start simulated annealing algorithm and an ant colony optimisation (ACO) algorithm to address the problem. Numerical results showed that these algorithms could produce high-quality solutions for instances with up to 200 nodes within reasonable computation times. Karimi (2018) explored the design of an advanced express system, incorporating both hub and vehicle capacity. They modelled the flow exchange of a warmed-up and capacitated express system for the first time and imposed maximum travel time constraints on origin-destination (OD) pairs. The design problem was formulated via a multi-commodity-flow approach and strengthened with four valid inequalities. Furthermore, they presented a tabu search (TS) algorithm as the solution method. Numerical tests demonstrated that the proposed valid inequalities effectively tightened the lower bounds, and the proposed TS algorithm successfully addressed the target problem.

The pickup and delivery process can be distinct due to logistical or scheduling considerations, as seen in the case of general freight forwarders. Sun (2015) studied a problem similar to that in Sun (2013), where the pickup and delivery processes were treated as separate. An endosymbiotic evolutionary algorithm was developed to simultaneously address hub location and vehicle routing, which was then evaluated on instances with 100 and 200 clients. The results indicated its applicability for supply-chain network planning. More recently, Yang et al. (2019) concentrated on the capacitated SAHLRP with distinct collection and distribution processes. They proposed a new mixed-integer linear programming model and developed an MA algorithm to tackle larger-sized problems, which could find high-quality solutions within acceptable computation times for benchmark instances.

Most studies have utilised heuristic algorithms (Ratli et al., 2022; Pandiri & Singh, 2021), with only a few attempts to solve the problem exactly. de Camargo et al. (2013) introduced a new SAHLRP model with simultaneous collections and distributions, accounting for the fixed costs of hubs and vehicles. They decomposed the problem into two subproblems: a transportation problem and a feasibility problem, and employed a tailored Benders decomposition algorithm to solve it optimally. Numerical results demonstrated that the algorithm could find optimal solutions for instances with up to 100 clients. Later, Rodriguez-Martin et al. (2014) investigated a variant of SAHLRP for an express system. In their problem, a circular path connected a predefined number of hubs, and the number of clients allocated to each hub was limited. Additionally, each hub and the clients assigned to it were connected by a single local tour. The problem was solved on instances with up to 50 clients using a branch-and-cut algorithm. Wu et al. (2024) recently developed a branch-and-price-and-cut algorithm to solve the capacitated SAHLRP on benchmark instances. Numerical results proved that the developed algorithm could efficiently solve the capacitated SAHLRP, outperforming the CPLEX solver.

All the studies above focused on the deterministic HLRPs, with only a few addressing stochastic SAHLRP. Mohammadi et al. (2013) investigated a multi-objective chance-constrained model for a stochastic green SAHLRP with time windows, incorporating stochastic travel and service time. To solve this problem, they introduced a multi-objective invasive weed optimisation algorithm and compared its performance with other state-of-the-art algorithms on randomly generated instances. In contrast, we focus on stochastic demands and employ a multi-stage recourse model, which sequentially integrates long-term and short-term decisions, making it more suitable for designing intra-city express systems. Musavi and Bozorgi-Amiri (2024) also addressed stochastic travel time and formulated the problem using a data-driven robust optimisation model. They constructed the uncertainty set via a support vector clustering method and provided numerical results on instances with 22 and 25 clients to demonstrate the significance of considering stochastic travel time. However, their model only considered the earliest possible delivery of parcels (i.e., delivering the parcels as soon as possible) instead of specifically required time windows, making the service less flexible. In contrast, we consider time windows, model the problem via a multi-stage recourse model, and conduct numerical experiments on significantly larger instances.

Later, Wang et al. (2023) developed a two-stage recourse model to deal with a hub location routing problem with direct transportation and stochastic demands. In their model, long-term decisions (hub location and client-to-hub allocation) were determined in the first stage, while short-term decisions (vehicle routing) were made in the second stage. However, they pointed out the difficulty of solving the recourse model and reformulated the problem as a distributionally robust model which simultaneously makes long-term and short-term decisions (without route failures and recourse policies). Then, they used a genetic algorithm with local search to solve the distributionally robust model, which was tested on a case study with 10 clients. As compared to Wang et al. (2023), our study focuses on the recourse model to sequentially integrate long-term decisions and short-term decisions, as the collection demands can only be revealed during the route execution. Therefore, our problem settings are more close to real operations. In addition, we incorporate time windows to improve service qualities and use an HSVNS algorithm to solve the model, which is tested on a series of much larger instances.

As reviewed above, the literature on stochastic HLRP is limited and based on simultaneously making long-term and short-term decisions. Our study is the first to model the HLRP with time windows and stochastic demands via the recourse model. Moreover, it introduces new recourse policies and proposes an HSVNS algorithm to solve the recourse model, making the first attempt to sequentially integrate long-term and short-term decisions in stochastic HLRPs.

3. Model formulation

The CHLRPTWSD is defined on a complete graph $G = (V, A)$, where V and A are vertex set and edge set, respectively. Vertex set V consists of potential hub set H and client (branch office) set C, while edge set A consists of edges between all vertices. For each pair of clients $i \in C$ and $j \in C$, d_{ij} denotes the flow to be transported from i to j through local tours and hubs, which is assumed to be a random variable with a known and independent distribution. Without loss of the generality, it is assumed that all realisations of d_{ij} are greater than 0 as well as that the collection demand ($O_i = \sum_{j \in C} d_{ij}$) and distribution demand ($D_i = \sum_{j \in C} d_{ji}$) of client $i \in C$ do not exceed the vehicle capacity (Che & Zhang, 2023). Each client $i \in C$ is addressed with a time window $[a_i, b_i]$ and a service time s_i, indicating its service start time should be no earlier than a_i and no later than b_i.

Each potential hub $k \in H$ has a capacity Q_k and a fixed cost F_k. In intra-city express systems, the flow collected from an origin client is initially sorted into flows with different destinations and then transported to the final destinations without further sorting operations (Ghaffarinasab, 2022). Therefore, we assume that only receiving parcel flows from clients consumes hub capacity, as described in Ernst and Krishnamoorthy (1999) and Hu et al. (2021). Local tours are operated by a large enough fleet of identical vehicles, each with a capacity q and a fixed cost f. Furthermore, backbone transportation is performed without

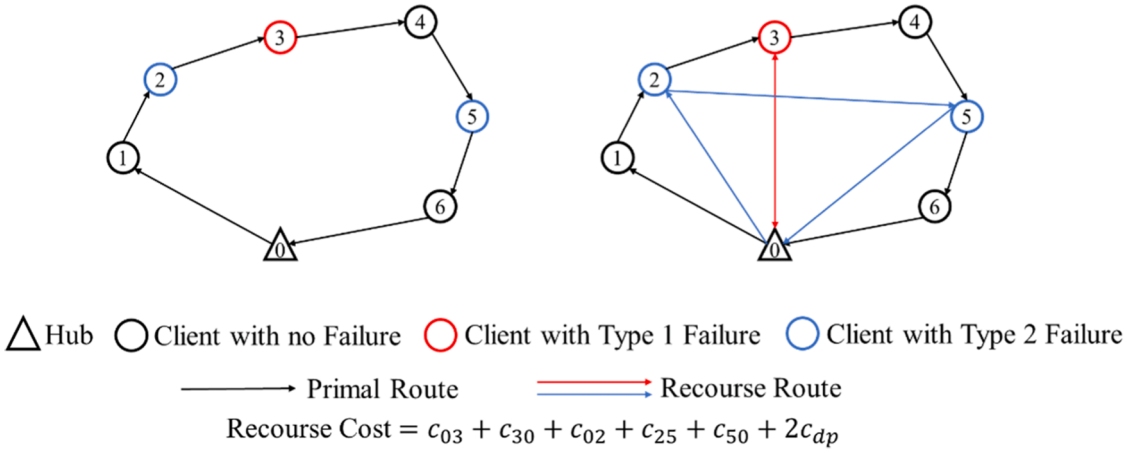


Fig. 4. Example of the recourse process.

capacity limitation and fixed cost, i.e., vehicle capacities are considered only in the local tour phase instead of the backbone transportation phase, and all hubs are fully interconnected.

Each edge $(i,j) \in A$ is addressed with a nonnegative travel distance c_{ij} and travel time t_{ij}, satisfying the triangle inequality. Local tour cost depends on the sum of travel distances of the used edges, while backbone transportation cost is calculated based on travel distances and transferred flows (Karimi, 2018; Wu et al., 2022). In addition, the unit backbone transportation cost (¥/km.t) and unit local tour cost (¥/km) are represented as α and β, respectively.

The CHLRPTWSD lies in stochastic programming, generally formulated by chance-constrained models and recourse models (Saint-Guillain et al., 2021). Based on the descriptions provided in Section 1, the CHLRPTWSD is formulated using a multi-stage recourse model as follows:

i) In the first stage, the hub location and the client-to-hub allocation (long-term decisions) are determined before any random variables ($d_{ij}|i,j \in C$) are realised.
ii) Then, in the second stage, the flows to be delivered to each client $i \in C$ ($d_{ji}|j \in C$) are revealed first (since these parcels have been collected in the previous service cycle, as shown in Section 1), forming the distribution demands ($D_i|i \in C$). Vehicle routing decisions (short-term decisions) are then made to link the hubs and clients before revealing the collection demands ($O_i|i \in C$).
iii) Finally, the collection demands are revealed in the third stage, and a predetermined recourse policy is applied when collection failures occur. We use the following recourse policy (Fig. 4): For any local tour $r = \{m,1,2,3,\ldots,|r|,m\}$, assuming a failure occurs at client $i \in \{1,2,3,\ldots,|r|\}$ due to its stochastic demand, if assigning a new vehicle from hub m to visit client i directly can start the service within its time window, then the vehicle satisfies client i's distribution demand, skips its collection demand, and continues the tour (Type 1 failure). The newly assigned vehicle will fulfil client i's collection demand; otherwise, the vehicle satisfies client i's distribution demand, skips its collection demand, pays a delay penalty c_{dp} and records it (Type 2 failure). After visiting all clients, another vehicle will traverse all recorded clients with their visiting sequence in r to complete the service. For simplicity, vehicle capacity constraints are excluded in Type 2 recourse actions since these skipped collection demands become known to the service providers, allowing them to use a sufficiently large vehicle to fulfil these demands in a single additional tour.

In other words, after long-tern decisions are made, a vehicle routing problem with simultaneous delivery and stochastic pickup (VRPSDSP) is solved for each installed hub and the clients assigned to it. Although these VRPSDSPs need to be solved multiple times for all the service cycles, we model them only once for simplicity, and the fixed costs are distributed across each service cycle to make long-term and short-term costs comparable. Furthermore, if the observed collection demand exceeds a hub's capacity due to uncertainty, a penalty cost, representing the overwork cost, must be paid. The unit overwork cost is expressed as ω. Note that one can set a higher ω to reduce the overwork possibility. Backbone transportation costs are also calculated in this stage.

For each edge $(i,j) \in A$, x_{ij} is a decision variable equal to 1 if there is a vehicle travelling directly from vertex i to vertex j, otherwise 0. $z_{ik}(i \in C, k \in H)$ is a decision variable equal to 1 if client i is allocated to hub k, otherwise 0. For each vertex $i \in V$, let u_i, v_i, and T_i be the expected collection load on the vehicle just after having served vertex i, the distribution load on the vehicle just after having served vertex i, and the service start time of vertex i, respectively. y_k is a decision variable equal to 1 if hub $k \in H$ is open, otherwise 0. Moreover, y_{ijkl} represents the fraction flow from client $i \in C$ to client $j \in C$ passing through hub $k \in H$ and hub $l \in H$. Finally, e_k represents the overwork load at hub $k \in H$, and Y is an enough large number.

The CHLRPTWSD is modelled as (1)-(25), in which $Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})$ and $Q_2(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})$ are the optimal values of the second-stage problem and third-stage problem, respectively. Random vector $\boldsymbol{\xi}$ contains flow d_{ij} to be transported from client $i \in C$ to $j \in C$. The notations are summarised in **Appendix A**.

$$Stage\ 1\ min \sum_{k \in H} F_k y_k + E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})] \tag{1}$$

$$s.t. \sum_{k \in H} z_{ik} = 1\ \forall i \in C \tag{2}$$

$$\sum_{i \in C} z_{ik} E[O_i] \le Q_k y_k\ \forall k \in H \tag{3}$$

$$\sum_{k \in H} z_{ik} t_{ik} \le b_i\ \forall i \in C \tag{4}$$

$$z_{ik} \in \{0,\ 1\}\ \forall i \in C,\ k \in H \tag{5}$$

$$b_k \in \{0,\ 1\}\ \forall k \in H \tag{6}$$

Objective function (1) minimises the operation cost, consisting of the hub fixed cost (the first term) and expected recourse cost incurred by the second-stage decisions (the second term), whose components can be found in objective function (7). Constraint (2) guarantees the single allocation between clients and hubs. Only open hubs can serve clients, as

Algorithm 1
HSVNS Algorithm.

Input: current solution S, best solution S_0, a set Opt of local optima, neighbourhood set $M = \{N_1, N_2, \ldots, N_{|M|}\}$
Initialize: generating initial solution $S_{initial}$, $S = S_0 = S_{initial}$, $Opt = Opt \cup S$
While terminal condition unsatisfied:
  Generate a sample Z (distribution demands)
  While $m \leq |M|$:
    Shaking: shake S in N_m and get a new solution S_{shake}
    Local Search: search for the best neighbour S' of S_{shake} on sample Z (Algorithm 2)
    Evaluation: evaluate S' via the ALNS algorithm (Section 4.1.2)
    If S' better than S_0
      $S_0 = S'$
      $S = S'$
      $Opt = Opt \cup \{S'\}$
    Endif
    If S' better than S
      $S = S'$
      $m = 1$
    Else
      Decide whether or not to accept S' via a SA procedure
      $m = m + 1$
    Endif
  End while
End while
Tournament: Generate a larger sample Z' and select the best S_0 on sample Z' in *Opt* (Algorithm 3)
Output S_0
End

Constraint (3) ensures. Also, this constraint guarantees that the expected collection load of a hub cannot exceed its capacity, thereby reducing overwork possibility. Constraint (4) guarantees the feasibility regarding the time window constraints of the local tours generated in the second stage. Constraints (5) and (6) are variable domains.

$$Stage\ 2\ Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi}) = \min \sum_{k\in H}\sum_{j\in V} f x_{kj} + \sum_{i\in V}\sum_{j\in V} \beta c_{ij} x_{ij} + E[Q_2(\boldsymbol{y},\boldsymbol{z},\boldsymbol{x},\boldsymbol{\xi})] \quad (7)$$

$$s.t. \sum_{j\neq i\in V} x_{ij} = 1\ \forall i \in C \quad (8)$$

$$\sum_{j\in V} x_{ji} = \sum_{j\in V} x_{ij}\ \forall i \in V \quad (9)$$

$$x_{ik} \leq z_{ik}\ \forall i \in C, k \in H \quad (10)$$

$$x_{ki} \leq z_{ik}\ \forall i \in C, k \in H \quad (11)$$

$$x_{ij} + z_{ik} + \sum_{l\neq k\in H} z_{jl} \leq 2\ \forall i \in C, j \neq i \in C, k \in H \quad (12)$$

$$u_i + E[O_j] + q(1 - x_{ij}) \leq u_j\ \forall i \in V, j \neq i \in C \quad (13)$$

$$v_i - D_j + q(1 - x_{ij}) \geq v_j\ \forall i \in V, j \neq i \in C \quad (14)$$

$$u_i + v_i \leq q\ \forall i \in C \quad (15)$$

$$T_i + t_{ij} + s_i + Y(1 - x_{ij}) \leq T_j\ \forall i \in V, j \neq i \in C \quad (16)$$

$$a_i \leq T_i \leq b_i\ \forall i \in C \quad (17)$$

$$x_{ij} \in \{0,1\}\ \forall i \in V, j \in V \quad (18)$$

$$u_i \geq 0\ v_i \geq 0\ T_i \geq 0\ \forall i \in V \quad (19)$$

Objective function (7) minimises the vehicle fixed cost (the first term), local tour cost (the second term), and expected recourse cost incurred by the third-stage decisions (the third term), whose components can be found in objective function (20). Each client should be visited exactly once unless collection failure occurs, which is guaranteed by Constraint (8). Constraint (9) balances the vehicle flow at each vertex. Constraints (10)-(12) link the allocation variables with routing variables. Constraints (13) and (14) describe the expected collection loads and distribution loads on vehicles. Note that collection and distribution are conducted simultaneously along local tours as well as that expected collection loads are considered to reduce collection failures, as in Sarasola et al. (2016). Vehicle capacity constraints are taken into account by Constraint (15). Consistency of service start time is ensured by Constraint (16), followed by Constraint (17) addressing time window constraints. Decision variables are defined by Constraints (18) and (19).

$$Stage\ 3\ Q_2(\boldsymbol{y},\boldsymbol{z},\boldsymbol{x},\boldsymbol{\xi}) = min\ R(\boldsymbol{x},\boldsymbol{\xi}) + \sum_{k\in H} \omega e_k + \sum_{i\in C}\sum_{j\in C}\sum_{k\in H}\sum_{l\in H} \alpha d_{ij} c_{kl} y_{ijkl} \quad (20)$$

$$s.t. \sum_{l\in H} y_{ijkl} = z_{ik}\ \forall i \in C,\ j \in C, k \in H \quad (21)$$

$$\sum_{k\in H} y_{ijkl} = z_{jl}\ \forall i \in C,\ j \in C, l \in H \quad (22)$$

$$e_k \geq \sum_{i\in C}\sum_{j\in C}\sum_{l\in H} d_{ij} y_{ijkl} - Q_k\ \forall k \in H \quad (23)$$

$$0 \leq y_{ijkl} \leq 1\ \forall i \in C, j \in C, k \in H, l \in H \quad (24)$$

$$e_k \geq 0\ \forall k \in H \quad (25)$$

Objective function (20) optimises the realised recourse cost, overwork cost, and backbone transportation cost, corresponding to the first term through the third term, respectively. In this context, $R(\boldsymbol{x},\boldsymbol{\xi})$ (the realised recourse cost) represents the cost associated with handling collection failures according to the recourse policy. Flow variables and allocation variables are correlated via Constraints (21) and (22). Overwork cost for each hub $k \in H$ is calculated by Constraint (23). Constraints (24) and (25) are variable domains. Since there is no simple way to model $R(\boldsymbol{x},\xi)$ via decision variables and linear relationships (Laporte et al., 2002), its specific formulation is not provided here. However, the method of calculating its expectations is presented in **Appendix B**.

4. Solution algorithm

The key to solving model (1)-(25) is calculating $E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})]$, which is very difficult even under a discrete distribution. More complicatedly, solving the first-stage problem (making long-term decisions) and the second-stage problem (making short-term decisions and calculating $E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})]$) are both NP-hard since they belong to hub location problems and vehicle routing problems, respectively, making the CHLRPTWSD NP-hard. As a result, we propose an HSVNS scheme to solve the problem. The SVNS algorithm, proposed by Gutjahr et al. (2007), follows the basic principles of the variable neighbourhood search algorithm but uses stochastic information to evaluate solutions by generating samples of random variables. Our HSVNS scheme embeds an ALNS algorithm into the SVNS framework and employs an SA procedure to avoid falling into local optima, as shown in Algorithm 1. The main modifications include: (i) The tournament procedure is implemented at the end of the algorithm to reduce computation time; (ii) The acceptance of inferior solutions helps escape from local optima. The effectiveness of these two modifications is evaluated in **Appendix C**.

4.1. Solution space and solution representation

Based on the problem descriptions, long-term decisions, i.e., $(\boldsymbol{y},\boldsymbol{z})$, are optimised via an SVNS algorithm, whereas short-term decisions (vehicle routing, i.e., $\boldsymbol{x}$) are optimised via an ALNS algorithm to obtain

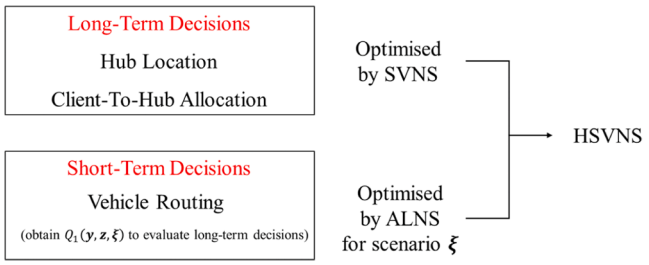


Fig. 5. HSVNS framework.

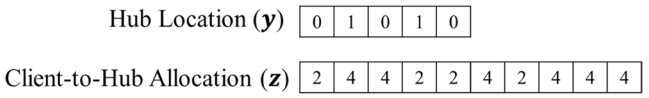


Fig. 6. An example of the solution representation.

$Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})$ for each scenario $\boldsymbol{\xi}$ to evaluate long-term decisions, forming the HSVNS algorithm, as shown in Fig. 5.

Therefore, a solution S is represented as a set of two arrays: the first array denotes the hub location, where value 1 in position k indicates that hub k is selected, otherwise 0; the second array includes the client-to-hub allocation information, where value k in position i means that client i is assigned to hub k. An example of the solution representation is illustrated in Fig. 6, where hubs 2 and 4 are selected, and clients 1,4,5,7 and 2,3,6,8,9,10 are assigned to hubs 2 and 4, respectively.

4.2. Objective value

4.2.1. Objective function

We allow the situation that the total expected collection demand assigned to a hub to exceed its capacity to extend the search space, i.e., the violation of Constraint (3). However, penalty cost must be added to the objective function to guide the search toward feasible solutions. More clearly, we use the following objective:

$$f(S) = \sum_{k\in H} F_k y_k + \mu v(S) + E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})] \tag{26}$$

where $v(S)$ is the violation of hub capacity constraints, and μ denotes its unit penalty cost. μ is adjusted along the search process: if S is infeasible, μ is multiplied by $\psi > 1$; otherwise, it is divided by ψ. As previously mentioned, computing $E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})]$ is NP-hard, even for a fixed $(\boldsymbol{y},\boldsymbol{z})$ and a discrete distribution. Consequently, for a given $(\boldsymbol{y},\boldsymbol{z})$, $E[Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})]$ is approximated via a sampling method on sample Z (in search procedure) and sample Z' (in tournament procedure), i.e., via $\sum_{\xi\in Z} Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})/|Z|$ and $\sum_{\xi\in Z'} Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})/|Z'|$. For each $\boldsymbol{\xi}\in Z$ (or $\boldsymbol{\xi}\in Z'$), $Q_1(\boldsymbol{y},\boldsymbol{z},\boldsymbol{\xi})$, representing short-term costs in scenario $\boldsymbol{\xi}$, is calculated using the ALNS algorithm. Our ALNS algorithm is adapted from the one in Wu et al. (2022) (for the vehicle routing subproblem), whose modifications will be presented in the following subsection.

Algorithm 2
Greedy NN algorithm.

Input: long-term decisions $S = (\boldsymbol{y},\boldsymbol{z})$
  For each hub k whose $y_k = 1$:
    Select all clients i whose $z_{ik} = 1$ and form set C_k
    While C_k is not empty:
      Set current vertex $v = k$
      Set current expected residual pickup capacity $p = q$
      Set current residual delivery capacity $d = q$
      Set current service start time $t = 0$
      Initialise a new route r
      While feasible unlinked client remained:
        Select feasible unlinked client i that is nearest to v
        $t = max\{t + s_v + t_{vi}, a_i\}$
        $d = \min\{d - D_i, p - E[O_i]\}, p = p - E[O_i]$
        $v = i$, add i to r, remove i from C_k
      End While
      Add r to the initial solution
    End While
  End For
  Output generated initial solution
  End

4.2.2. Adaptive large neighbourhood search

a. Initial Solution Generation

The following greedy NN algorithm (Algorithm 2) is used to generate initial vehicle routing for a given solution $S = (\boldsymbol{y},\boldsymbol{z})$ as the start of the ALNS algorithm, where feasible unlinked clients are defined as those clients i satisfying (i) $i \in C_k$ (ii) $p \geq E[O_i]$ (iii) $d \geq D_i$ (iv) $t + s_v + t_{vi} \leq b_i$. b. Destroy/Repair Methods

Highest COV Removal: Clients with high coefficient-of-variation (COV) demands are selected and inserted into the client pool. The COVs are randomised and normalised to expand the search space.

Largest Time Gap Removal: This method removes some clients whose vehicle arrival times are earlier than their corresponding early time windows or later than their corresponding late time windows. This method aims to make use of the working horizon of vehicles.

Both **Greedy Insertion** and **Regret Greedy Insertion** are used as repair methods. When applying repair methods, time windows are considered in accordance with Eq. (26), i.e., violating time window constraints is allowed but should be penalised. In **Greedy Insertion**, the clients are inserted into the solution randomly at the place with minimum insertion cost, in which the insertion cost is the difference between the total cost after the considered client is inserted and the total cost before it is inserted. While calculating the insertion cost of a client at a particular position in the route r, the "push forward" effect is considered, and the route r is checked for the time window feasibility for clients following the inserted client. In Regret Greedy Insertion, a regret is defined for each client as the difference between the insertion cost of the best and the second-best insertion, and the clients with higher regret value will be prioritised for insertion. Note that the hub location and client-to-hub allocation remain unchanged.

4.3. Shaking

Four shaking neighbourhood structures are utilised: **Reallocation** (N_1), **Allocation Exchange** (N_2), **One-Hub-Close** (N_3), and **One-Hub-Open** (N_4).

Reallocation (N_1)**:** This neighbourhood structure includes all solutions obtained by reallocating one client from its current hub to a new one.

Allocation Exchange (N_2)**:** This neighbourhood structure includes all solutions obtained by swapping the client-to-hub relationship of two clients.

One-Hub-Close (N_3)**:** This neighbourhood structure includes all solutions obtained by closing one installed hub. The clients allocated to the closed hub are randomly reallocated to other installed hubs.

One-Hub-Open (N_4)**:** This neighbourhood structure includes all solutions obtained by installing one closed hub. Some clients are randomly selected, removed from their current hubs, and reallocated to the installed hub.

4.4. Local search

4.4.1. Neighbourhood structures

Local search is performed in the following four neighbourhood structures: **Reallocation** (N_1), **Two-Clients-Shaw-Swap** (N_5), **Two-Hubs-Swap** (N_6), and **Close-and-Open** (N_7), presented in Figs. 7–9, respectively. The first three neighbourhoods optimise the allocation between clients and hubs, while the last one searches for a better hub

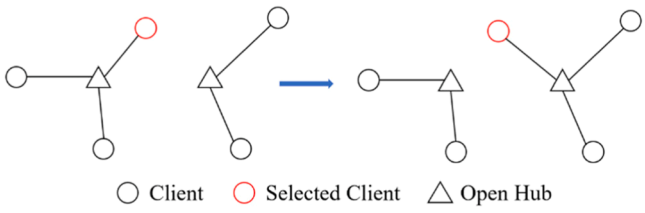


Fig. 7. Example of N_1.

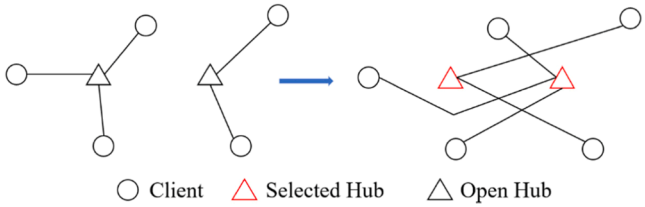


Fig. 8. Example of N_6.

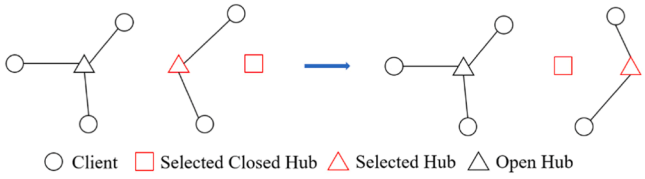


Fig. 9. Example of N_7.

backbone.

Reallocation (N_1)**:** This neighbourhood structure includes all solutions obtained by reallocating one client from its current hub to a new one. Note that if the reallocated client is the only client allocated to hub m, then hub m is closed. This process is illustrated in Fig. 7.

Two-Clients-Shaw-Swap (N_5)**:** This neighbourhood structure swaps the client-to-hub relationship of one client with its most related client. The relatedness R_{ij} of two clients i and j is defined below, in which hub k and l are current hubs of clients i and j, and $\omega_1 - \omega_3$ are Shaw parameters:

$$R_{ij} = \omega_1 \frac{min(c_{ik}, c_{jl})}{max(c_{ik}, c_{jl})} + \omega_2 \left(\frac{d_{ij}}{O_i} + \frac{d_{ji}}{D_i}\right) + \omega_3 \frac{min(b_i, b_j) - max(a_i, a_j)}{max(b_i, b_j) - min(a_i, a_j)} \quad (27)$$

Two-Hubs-Swap (N_6): Two open hubs k and l are selected, and the clients allocated to k are reallocated to l (and vice versa), as shown in Fig. 8. Note that the hub backbone remains unchanged.

Close-and-Open (N_7): The hub backbone is optimised in this neighbourhood structure by closing one open hub k and then opening one closed hub l. Then, the clients allocated to hub k are reallocated to hub l. An example of this process is illustrated in Fig. 9.

Algorithm 3
SVND Algorithm.

Input: shaked solution S_{shake}, set of neighbourhoods $N_m = \{N_1, N_5, N_6, N_7$
  Initialise: $m = 1$
  While not all neighbourhoods explored:
    Find the first S' in $N_m(S_{shake})$ better than (S_{shake})
    if S' exists:
      $(S_{shake}) = S'$
      Return N_1 and update m
    Else
      Move to the next neighbourhood and update m
    Endif
  End while
  OutputS'
  End

4.4.2. Sequential variable neighbourhood descent

The neighbourhoods are explored one by one in the given order (N_1, N_5,N_6,N_7) based on the increasing cardinality, resulting in a sequentially variable neighbourhood descent (SVND, Ilić et al., 2010) scheme (see Algorithm 3). The search goes to the next neighbourhood structure if it falls into the local optima. Otherwise, it returns to N_1.

4.5. Simulated annealing procedure

An SA procedure accepts inferior solutions to escape from local optima. Note that this procedure is executed only at the end of each iteration but not in the local search stage. The probability of accepting an inferior solution s' $(f(S') > f(S))$ is calculated as follows:

$$P = e^{-(f(S') - f(S))/T} \quad (28)$$

where T, referred to as the temperature, is set and updated following the method described by Ropke and Pisinger (2006).

4.6. Tournament

The tournament procedure outlined in Algorithm 4 is conducted in a two-round style. In the first round, the first half S_0 with lower $f(S_0|Z)$ is selected to form Opt' and compared using sample Z'. Then, in the second round, the first half S_0 with lower $f(S_0|Z')$ in Opt' is selected to form Opt'' and compared using sample Z' again because of the heuristic nature of the objective function calculation method (i.e., ALNS). Finally, the best S_0 is selected from Opt''. This process aims to balance the computation time and comparison accuracy.

5. Numerical expreiment

The proposed model and algorithm are evaluated on the instances adapted from the literature. We conduct the experiments in the manner described by Shi et al. (2018): i) The flows are set to be deterministic first, and the problem degenerates to a deterministic problem, i.e., capacitated hub location routing problem with time windows (CHLRPTW). The proposed algorithm is then compared with the CPLEX on solving the CHLRPTW to evaluate its performance. ii) The HSVNS algorithm is applied to solve the stochastic problem (CHLRPTWSD), and its solutions are compared with those of the deterministic problem to show the advantages of considering stochastic factors. iii) A sensitivity analysis of time window features is undertaken to gain insights into the CHLRPTWSD. Section 5.1 introduces the test instances. Algorithm parameters are provided in Section 5.2. The experiment results for the deterministic and stochastic problems are evaluated and compared in Sections 5.3 and 5.4, respectively, followed by Section 5.5 as the sensitivity analysis. We coded the algorithm in Java 7 and ran the experiments on a computer with i5–13600KF CPU and 32 GB RAM. Moreover, we use CPLEX 22.1.1 for comparison with the proposed algorithm in Section 5.3.

5.1. Instance generation

We test the proposed model and algorithm on instances with up to 70 clients. Instances with 10, 15, 20, 25, 30, and 40 clients are modified from those in Wu et al. (2024), while instances with 50, 60, and 70 clients are newly generated following Wu et al. (2024) (instances with modified flows). Each instance is associated with five or ten potential hubs randomly selected from the client set. All these instances have the same base dataset (Australia Post dataset), where those with {10, 15, 20, 25} clients, {30, 35, 40} clients, and {50, 60, 70} clients are categorised into small, medium, and large instances, respectively. The instances are labelled as $H-C-Q-F$, where H and C denotes the number of

Algorithm 4
Tournament.

Input: a set *Opt* of local optima
First Round:
Rank S_0 in *Opt* according to increasing $f(S_0|Z)$
Select the first half S_0 in *Opt* and form Opt'
Calculate $f(S_0|Z')$ for all S_0 in Opt'
Second Round:
Rank S_0 in Opt' according to increasing $f(S_0|Z')$
Select the first half S_0 in Opt' and form Opt''
Recalculate $f(S_0|Z')$ for all S_0 in Opt''
Select the best S_0 in Opt''
Output S_0
End

Table 1
Parameter setting.

Notation	Definition	Value
μ	Unit penalty cost on hub capacity violation	100
ψ	Penalty cost factor	1.05
$\omega_1, \omega_2, \omega_3$	Shaw parameters	0.3,0.3,0.4
Z	Sample size used in the search process	10
Z'	Sample size used in the tournament	100

potential hubs and clients, respectively, Q and F represent the feature (tight and loose) of hub capacity and fixed cost, respectively. For example, instance 5–15-L-T means an instance with 5 potential hubs and 15 clients, where the hub capacity and fixed cost are loose and tight, respectively.

The coordinates of the clients and potential hubs and the hub capacity and fixed costs are used directly. For each arc $(i,j)|i \in V\, j \in V$, c_{ij} and t_{ij} are set equal to the Euclidean distance between vertex i and j. Some additional data are added to complete the generated instances. The flow d_{ij} for each pair of clients i and j $(j \neq i)$ is assumed to follow a uniform distribution $[0.5\widehat{d}_{ij}, 1.5\widehat{d}_{ij}]$, in which the value of $\widehat{d}_{ij}$ is provided by the Australia Post generator. Service time is set as a fixed value of 2000, consistent with the Solomon benchmark (Solomon, 1987). Time windows are generated using the method described by Li et al. (2010):

i) For each client $i \in C$, its time window $[a_i, b_i]$ is defined by a time window centre c_i^{tw} and a time window width w_i^{tw}, in which $a_i = c_i^{tw} - 0.5w_i^{tw}$ and $b_i = c_i^{tw} + 0.5w_i^{tw}$.
ii) For each client $i \in C$, c_i^{tw} is randomly selected from the range of $[max(t_{ik}|k \in H),\ l_0 - max(t_{ik}|k \in H) - s_i]$, with l_0 is set to 150,000 for all instances. Moreover, w_i^{tw} is randomly selected from the interval of [20,000, 70,000].
iii) The following conditions must be satisfied for each client $i \in C$: $a_i \geq 0,\ b_i \geq min(t_{ik}|k \in H)$.

In addition, α and β are set to 0.002 and 1 to balance the backbone transportation cost and local tour cost, respectively, which are similar to Rodríguez-Martín et al. (2014). Finally, f was fixed at 3000 for all instances.

5.2. Parameter setting

We explore the HSVNS parameters in preliminary experiments and summarise their values in Table 1. The terminal conditions are set to a maximum number of iterations of 300, 450, and 600 for small, medium, and large instances, respectively. To calculate $Q_1(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\xi})$ for each scenario $\boldsymbol{\xi}$, the maximum number of iterations for the ALNS algorithm is set to 1000, 2000, and 3000 for small, medium, and large instances, respectively.

5.3. Computation results on the CHLRPTW

The proposed algorithm is compared with the CPLEX on small-sized and medium-sized CHLRPTW instances, whose results are shown in Table 2 and Table 3, respectively. The notation definitions are given below. We do not compare the CPLEX and the proposed algorithm on large instances since the CPLEX performs significantly worse in these instances.

UB: Best upper bound provided by the CPLEX within 10,800 s.

T: CPU time in seconds spent solving the problem using the corresponding method.

F_{best}: Best objective value obtained in 10 runs by the proposed algorithm.

F_{aver}: Average objective value obtained in 10 runs by the proposed algorithm.

Gap_{UB}: Deviation in % between F_{best} and *UB*: $Gap_{UB} = (UB - F_{best})/F_{best} \times 100\%$.

Gap_{aver}: Deviation in % between F_{best} and F_{aver}: $Gap_{aver} = (F_{aver} - F_{best})/F_{best} \times 100\%$.

COV: COV for the objective values in 10 runs calculated with F_{aver}.

Solution Quality. It can be observed in Tables 2 and 3 that the CPLEX provided optimal solutions in the given time only for instances with up to 20 clients. For larger instances, the solution quality deteriorated (e.g., instances with 40 clients). In contrast, the proposed algorithm yields high-quality solutions for all instances, including 22 optimal solutions ($Gap_{UB} = 0$ with zero optimality gaps), 5 same solutions ($Gap_{UB} = 0$ with positive optimality gaps), and 29 better solutions ($Gap_{UB} > 0$). For the 29 instances where the proposed algorithm found better solutions, Gap_{UB} varies between 0.17 % and 6.77 %. Furthermore, the relatively small Gap_{aver} (0.41 % on average and 1.90 % in worst case) and *COV* (0.45 % on average and 1.62 % at worst) shows the robustness of the proposed algorithm. One can also find that the proposed algorithm outperforms the CPLEX in solution quality, indicating its capability to provide high-quality solutions for the sampling problem. Moreover, the solutions obtained for the CHLRPTW are to be compared with the CHLRPTWSD solutions in Section 5.4.

Computation time. Compared to the CPLEX, our proposed algorithm can solve the CHLRPTW in substantially less computation time. For the instances exactly solved by the CPLEX, the computation times are critically decreased (from an average of 575.82 s to 54.53 s). Furthermore, the average computation times for the proposed algorithm are 219.81 s, 772.71 s, 1298.19 s, and 1831.83 s in instances with 25, 30, 35, and 40 clients, respectively.

5.4. Computation results on the CHLRPTWSD

In this section, the performance of the stochastic model is compared with that of the deterministic model. For the stochastic model, we employ the HSVNS framework ($|Z| = 10$, $|Z'| = 100$), in which samples Z and Z' are populated based on the distribution shown in Section 5.1. For the deterministic model, the solutions presented in Section 5.4 are used, in which the values of the random variables are set as their mathematical expectations. After solving both the stochastic and deterministic models, we generate a new sample with the size of 1000 (called the evaluation sample) to assess the qualities of these solutions. The comparison is summarised in Tables 4–6, with the definition of the notations presented below.

F_{det}: Objective value on the evaluation sample of the solution for the deterministic model.

Hub: Installed hubs in the solution for the corresponding model.

F_{stc}: Objective value on the evaluation sample of the solution for the stochastic model.

Gap_{det}: Deviation in % between F_{stc} and F_{det}: $Gap_{det} = (F_{det} - F_{stc})/F_{stc} \times 100\%$.

Computation time. Columns "*T*" indicates that the HSVNS framework can solve the problem within acceptable computation times, with

Table 2
Computation results on small-sized CHLRPTW instances.

	CPLEX		Proposed Algorithm					
Instance	*UB*	*T*(*s*)	F_{best}	F_{aver}	Gap_{UB}(%)	Gap_{aver}(%)	*COV*(%)	*T*(*s*)
5–10-L-L	226,243.92	0.59	226,243.92	226,243.92	0	0	0	11.42
5–10-L-T	240,438.79	0.89	240,438.79	240,438.79	0	0	0	11.76
5–10-T-L	252,184.86	0.48	252,184.86	252,184.86	0	0	0	17.22
5–10-T-T	277,666.79	0.58	277,666.79	277,666.79	0	0	0	16.12
10–10-L-L	240,867.61	11.52	240,867.61	240,867.61	0	0	0	11.80
10–10-L-T	254,668.86	13.31	254,668.86	254,668.86	0	0	0	11.15
10–10-T-L	284,333.45	2.03	284,333.45	284,333.45	0	0	0	18.82
10–10-T-T	295,786.69	2.19	295,786.69	295,786.69	0	0	0	15.80
5–15-L-L	304,649.75	158.88	304,649.75	307,174.52	0	0.83	1.26	40.17
5–15-L-T	338,369.90	150.84	338,369.90	338,369.90	0	0	0	44.43
5–15-T-L	355,483.45	108.12	355,483.45	355,483.45	0	0	0	68.64
5–15-T-T	402,639.16	106.98	402,639.16	402,639.16	0	0	0	64.97
10–15-L-L	290,281.02	1021.97	290,281.02	291,771.06	0	0.51	0.78	48.48
10–15-L-T	304,906.96	564.30	304,906.96	304,906.96	0	0.01	0.04	47.50
10–15-T-L	348,131.90	251.70	348,131.90	348,210.28	0	0.02	0.04	75.06
10–15-T-T	358,446.66	351.28	358,446.66	358,446.66	0	0	0	73.26
5–20-L-L	350,463.76	2455.86	350,463.76	351,760.42	0	0.37	0.82	79.19
5–20-L-T	380,198.75	1082.64	380,198.75	380,198.75	0	0	0	69.90
5–20-T-L	399,615.91	1792.38	399,615.91	399,636.42	0	0.01	0.01	140.35
5–20-T-T	428,523.17	393.12	428,523.17	428,523.17	0	0	0	165.08
10–20-L-L	311,214.70	2847.92	311,214.70	315,107.93	0	0.72	1.25	70.55
10–20-L-T	317,670.56	1350.52	317,670.56	321,674.37	0	0.85	1.26	98.03
10–20-T-L	380,959.51	10,800	380,959.51	386,320.00	0	0.54	1.41	114.32
10–20-T-T	395,266.86	10,800	395,266.86	398,678.03	0	0.56	0.86	102.55
5–25-L-L	390,119.38	10,800	371,627.87	371,627.87	4.98	0	0	247.86
5–25-L-T	425,666.44	10,800	398,692.93	398,692.93	6.77	0	0	208.56
5–25-T-L	428,904.31	10,800	428,196.10	428,886.44	0.17	0.16	0.30	252.25
5–25-T-T	476,360.30	10,800	462,274.62	464,378.19	3.05	0.46	0.39	250.72
10–25-L-L	359,379.65	10,800	358,768.19	359,166.64	0.17	0.11	0.20	169.16
10–25-L-T	371,568.26	10,800	367,756.02	368,585.67	1.04	0.23	0.50	200.40
10–25-T-L	403,300.35	10,800	398,738.30	402,897.57	1.14	1.04	1.32	202.44
10–25-T-T	417,621.25	10,800	413,059.20	417,990.44	1.10	1.19	1.36	227.07

Table 3
Computation results on medium-sized CHLRPTW instances.

	CPLEX		Proposed Algorithm					
Instance	*UB*	*T*(*s*)	F_{best}	F_{aver}	Gap_{UB}(%)	Gap_{aver}(%)	*COV*(%)	*T*(*s*)
5–30-L-L	452,788.37	10,800	448,894.22	449,416.94	0.87	0.12	0.24	802.06
5–30-L-T	454,331.02	10,800	454,331.02	456,157.25	0	0.40	0.41	863.39
5–30-T-L	458,904.05	10,800	454,129.55	458,089.13	1.05	0.87	0.81	665.51
5–30-T-T	465,725.22	10,800	462,494.87	465,133.59	0.70	0.57	0.60	712.75
10–30-L-L	432,637.98	10,800	432,637.98	433,286.94	0	0.15	0.29	834.26
10–30-L-T	445,097.01	10,800	445,097.01	447,010.93	0	0.43	0.51	896.22
10–30-T-L	483,720.73	10,800	475,261.08	479,348.33	1.78	0.86	0.84	711.67
10–30-T-T	499,324.67	10,800	489,581.99	492,715.31	1.99	0.64	0.71	695.83
5–35-L-L	470,069.16	10,800	465,962.87	467,607.12	0.88	0.35	0.31	1269.85
5–35-L-T	492,154.28	10,800	474,617.10	475,192.33	3.70	0.12	0.13	1292.85
5–35-T-L	494,277.14	10,800	475,150.13	482,885.76	4.03	1.63	0.88	1303.88
5–35-T-T	509,918.27	10,800	483,515.45	491,631.76	5.46	1.68	1.62	1368.04
10–35-L-L	470,144.45	10,800	464,937.15	466,703.91	1.12	0.38	0.32	1302.77
10–35-L-T	489,909.82	10,800	470,253.23	471,193.74	4.18	0.20	0.24	1368.93
10–35-T-L	502,532.58	10,800	487,517.05	494,439.79	3.08	1.42	0.81	1182.46
10–35-T-T	526,969.77	10,800	502,977.73	512,534.31	4.77	1.90	1.59	1296.73
5–40-L-L	530,481.82	10,800	515,098.66	518,982.35	2.99	0.75	0.47	1822.63
5–40-L-T	547,010.15	10,800	523,463.98	525,444.77	4.50	0.38	0.34	1830.49
5–40-T-L	565,115.02	10,800	529,900.01	529,989.30	6.65	0.02	0.03	1700.67
5–40-T-T	571,721.02	10,800	546,271.08	546,315.73	4.66	0.01	0.02	1669.92
10–40-L-L	526,200.53	10,800	511,470.19	515,101.63	2.88	0.71	0.56	1962.36
10–40-L-T	536,905.62	10,800	512,754.87	517,984.97	4.71	1.02	0.47	1977.84
10–40-T-L	551,537.76	10,800	525,174.02	528,114.99	5.02	0.56	0.42	1861.31
10–40-T-T	560,201.93	10,800	535,463.52	537,658.92	4.62	0.41	0.50	1829.41

average computation times of 387.15 s, 6921.04 s, and 26,145.98 s for small, medium, and large instances, respectively. These computation times are acceptable as long-term decisions should be determined only once for each network. In contrast, short-term decisions can be made quickly for each service cycle. Finally, one can find that the computation times for the stochastic model are significantly longer than those for the deterministic model due to the multiple scenarios in Z and Z'.

Cost Reduction. The tables show that incorporating stochastic factors could efficiently reduce operation costs, with an average reduction of 8.11 % and a maximum reduction of 49.48 %. Specifically, the average cuts are 15.33 %, 3.36 %, and 3.24 % for small, medium, and large instances. The reduction diminishes as the instance size increases

Table 4
Computation results on small-sized CHLRPTWSD instances.

	CHLRPTW			CHLRPTWSD			
Instance	F_{det}	*Hub*	$T(s)$	F_{stc}	$Gap_{det}(\%)$	*Hub*	$T(s)$
5–10-L-L	349,166.22	5	11.42	233,584.18	49.48	5	42.03
5–10-L-T	356,630.80	2,4	11.76	248,821.02	43.33	2,4	42.27
5–10-T-L	347,828.05	1,2,4	17.22	260,468.58	33.54	4,5	44.69
5–10-T-T	373,271.84	2,3,4	16.12	280,315.83	33.16	2,3,4	48.54
10–10-L-L	374,175.68	5,7	11.80	250,603.23	49.31	5,7	43.51
10–10-L-T	380,158.39	4,5	11.15	263,852.30	44.08	4,5	47.82
10–10-T-L	390,532.63	5,9,10	18.82	291,747.07	33.86	5,9,10	55.73
10–10-T-T	402,723.81	5,9,10	15.80	302,390.61	33.18	5,9,10	50.69
5–15-L-L	368,679.85	4,5	40.17	330,328.64	11.61	4,5	77.90
5–15-L-T	387,115.52	1,3	44.43	367,286.58	5.40	1,3	109.63
5–15-T-L	389,998.30	1,2,5	68.64	383,467.07	1.70	1,2,5	95.30
5–15-T-T	447,532.62	1,4	64.97	432,613.16	3.45	1,4	148.69
10–15-L-L	361,358.46	5,7	48.48	325,929.88	10.87	5,7	91.15
10–15-L-T	371,473.19	5,10	47.50	346,232.82	7.29	5,10	110.34
10–15-T-L	392,521.12	4,5,10	75.06	383,321.41	2.40	4,5,10	145.81
10–15-T-T	401,741.01	4,5,10	73.26	392,823.91	2.27	4,5,10	164.53
5–20-L-L	392,210.97	4,5	79.19	364,769.89	7.52	4,5	427.62
5–20-L-T	403,315.67	2,4	69.90	394,621.29	2.20	2,4	413.57
5–20-T-L	476,987.34	2,4,5	140.35	414,097.09	15.19	2,4,5	352.88
5–20-T-T	463,007.38	2,3,4	165.08	441,234.32	4.93	2,3,4	351.99
10–20-L-L	358,613.95	6,9	70.55	334,777.77	7.12	6,9	462.19
10–20-L-T	364,200.91	6,9	98.03	340,342.87	7.01	6,9	416.76
10–20-T-L	456,549.75	6,9,10	114.32	401,327.14	13.76	6,9,10	375.14
10–20-T-T	472,228.36	6,9,10	102.55	415,584.23	13.63	6,9,10	369.35
5–25-L-L	422,562.59	2,5	247.86	381,458.69	10.78	2,5	972.63
5–25-L-T	424,745.10	2,3	208.56	420,291.42	1.06	2,3	1011.20
5–25-T-L	464,226.55	2,4,5	252.25	449,220.05	3.34	2,4,5	893.70
5–25-T-T	521,048.75	2,3,4	250.72	483,137.46	7.85	2,3,4	908.10
10–25-L-L	400,577.04	6,8	169.16	365,023.73	9.74	6,8	1045.68
10–25-L-T	412,150.39	5,6	200.40	376,358.68	9.51	5,6	1102.39
10–25-T-L	448,713.51	5,6,10	202.44	423,115.05	6.05	5,6,10	995.86
10–25-T-T	463,016.34	5,6,10	227.07	437,633.59	5.80	5,6,10	971.26

due to the vehicle capacity constraint becoming less strict, i.e., the solutions for the deterministic and stochastic models tend to coincide when the instance size becomes larger. The average reductions for instances with different features are shown in Fig. 10, from which one can see that the reductions are notably more significant for instances with loose hub capacity and fixed cost. This can be attributed to the following reasons: i) for instances with loose hub capacity, each vehicle serves more clients compared to those with tight hub capacity; ii) local tour cost constituted a more significant portion for instances with loose fixed cost compared to those with tight fixed cost.

Hubs and Client-to-Hub Allocation. In most instances, the same hubs are installed in solutions for both the deterministic and stochastic models, except 5–10-T-L and 5–30-L-T. Since short-term decisions vary with the realisations of random variables, we focus only on long-term

Table 5
Computation results on medium-sized CHLRPTWSD instances.

	CHLRPTW			CHLRPTWSD			
Instance	F_{det}	*Hub*	$T(s)$	F_{stc}	$Gap_{det}(\%)$	*Hub*	$T(s)$
5–30-L-L	481,002.65	3,4	802.06	463,866.40	3.69	3,4	4271.81
5–30-L-T	497,465.11	4	863.39	467,017.18	6.52	3,4	3970.32
5–30-T-L	480,439.91	2,4	665.51	466,419.54	3.01	2,4	3611.55
5–30-T-T	484,578.33	2,4	712.75	472,810.72	2.49	2,4	3310.38
10–30-L-L	477,837.76	8	834.26	459,371.04	4.02	8	4381.26
10–30-L-T	489,374.13	4,6	896.22	470,823.68	3.94	4,6	4151.33
10–30-T-L	475,863.22	5,6,10	711.67	461,510.25	3.11	5,6,10	3892.05
10–30-T-T	490,278.65	5,6,10	695.83	475,906.28	3.02	5,6,10	3448.19
5–35-L-L	508,231.10	3,4	1269.85	485,756.26	4.63	3,4	6535.57
5–35-L-T	516,133.22	3,4	1292.85	493,586.83	4.57	3,4	7253.27
5–35-T-L	518,718.96	2,4	1303.88	493,959.32	5.01	2,4	6418.26
5–35-T-T	526,239.12	2,4	1368.04	499,839.42	5.28	2,4	6352.26
10–35-L-L	507,813.37	5,6	1302.77	486,178.43	4.45	5,6	6715.09
10–35-L-T	513,074.07	5,6	1368.93	491,403.19	4.41	5,6	7141.34
10–35-T-L	541,714.30	6,9,10	1182.46	512,647.20	5.67	6,9,10	6628.82
10–35-T-T	557,374.48	6,9,10	1296.73	527,816.74	5.60	6,9,10	6123.68
5–40-L-L	529,911.58	2,4	1822.63	521,147.08	1.68	2,4	9707.13
5–40-L-T	539,587.46	2,4	1830.49	531,549.08	1.51	2,4	10,523.91
5–40-T-L	544,326.73	2,3,4	1700.67	538,683.13	1.05	2,3,4	10,401.83
5–40-T-T	558,987.27	2,3,4	1669.92	553,868.87	0.92	2,3,4	10,014.85
10–40-L-L	534,506.75	4,9	1962.36	523,974.86	2.01	4,9	10,302.01
10–40-L-T	535,623.36	4,9	1977.84	525,120.94	2.00	4,9	10,628.21
10–40-T-L	547,484.34	4,9,10	1861.31	541,741.88	1.06	4,9,10	10,005.09
10–40-T-T	557,134.93	4,9,10	1829.41	552,110.72	0.91	4,9,10	10,316.74

Table 6
Computation results on large-sized CHLRPTWSD instances.

	CHLRPTW			CHLRPTWSD			
Instance	F_{det}	*Hub*	$T(s)$	F_{stc}	$Gap_{det}(\%)$	*Hub*	$T(s)$
5–50-L-L	614,737.90	2,4	2336.60	578,901.02	6.19	2,4	17,272.71
5–50-L-T	625,819.75	2,4	2456.08	583,986.35	7.16	2,4	17,863.74
5–50-T-L	605,340.21	1,2,4	2448.52	592,671.71	2.14	1,2,4	20,069.76
5–50-T-T	638,264.17	2,3,4	2385.63	611,395.77	4.39	2,3,4	19,953.41
10–50-L-L	604,992.54	5,8	2417.13	570,156.01	6.11	5,8	18,402.36
10–50-L-T	621,547.31	5,8	2526.92	585,923.18	6.08	5,8	17,841.69
10–50-T-L	601,123.08	4,7,10	2415.66	589,741.08	1.93	4,7,10	20,946.02
10–50-T-T	632,325.35	4,6,10	2298.57	607,012.91	4.17	4,6,10	19,726.51
5–60-L-L	682,326.89	2,5	3354.61	651,663.11	4.71	2,5	22,500.54
5–60-L-T	684,568.16	1,2	3471.87	677,993.06	0.97	1,2	23,028.77
5–60-T-L	682,562.23	1,2,5	3518.27	670,892.67	1.74	1,2,5	24,608.64
5–60-T-T	762,332.37	1,2,5	3676.84	756,591.55	0.76	1,2,5	24,095.62
10–60-L-L	674,392.88	3,7	3459.11	647,520.77	4.15	3,7	23,187.26
10–60-L-T	687,146.37	5,6	3486.78	678,596.06	1.26	5,6	22,547.38
10–60-T-L	676,212.52	4,7,8	3215.63	665,891.21	1.55	4,7,8	24,811.06
10–60-T-T	706,182.22	4,5,6,8	3374.16	701,482.29	0.67	4,5,6,8	24,173.31
5–70-L-L	706,070.76	1,3	5165.97	694,342.51	1.69	1,3	35,256.17
5–70-L-T	714,637.35	3,4	5637.21	677,460.27	5.49	3,4	35,841.28
5–70-T-L	704,971.27	1,3,4	5863.56	687,736.58	2.51	1,3,4	36,112.90
5–70-T-T	730,836.23	1,3,4	5771.25	714,066.23	2.35	1,3,4	35,873.79
10–70-L-L	703,854.80	4,5	5420.03	691,273.62	1.82	4,5	36,287.11
10–70-L-T	731,999.21	5,6	5697.60	691,870.71	5.80	5,6	35,973.45
10–70-T-L	719,015.01	4,5,6,10	5391.45	704,157.29	2.11	4,5,6,10	36,082.10
10–70-T-T	743,451.39	4,5,6,10	5489.64	728,302.69	2.08	4,5,6,10	35,048.03

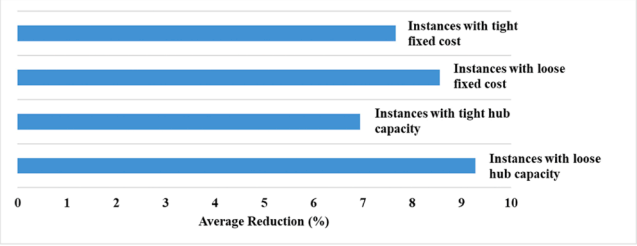


Fig. 10. Average reductions for instances with different features.

ones for the deterministic and stochastic models on instance 5–10-T-L, as illustrated in Fig. 11. In the deterministic model, one vehicle serves clients 2, 4, and 6, while they are served separately in the stochastic model, resulting in a different hub backbone. In other instances, client-to-hub allocation may be different, even if the same hubs are installed, as shown in Fig. 12. Clients 2, 4, and 6 cannot be served by a single vehicle due to stochastic demands. Therefore, clients 2 and 4 are assigned to hub 1 in the stochastic model rather than hub 3 in the deterministic model.

5.5. Sensitivity analysis

In practical scenarios, some clients may not have specific time windows. To investigate its impact on the CHLRPTWSD, we remove time windows for some randomly selected clients and conduct a sensitivity analysis of the time-window-free rate (TWFR). The analysis is conducted on instances 5–20 and 5–25. The results are shown in Tables 7 and 8. Moreover, the average reductions for instances with different TWFRs are illustrated in Fig. 13. It can be concluded from the figure that the average reduction increases with the TWFR. This phenomenon can be explained by each vehicle serving more clients in instances with a high TWFR, heightening the likelihood of collection failure. The sensitivity analysis results and the results in Section 5.4 indicate that considering stochastic factors can significantly reduce operation costs, especially for instances with tight vehicle capacity and high TWFR.

6. Case study

We generate three case studies using a domestic population distribution dataset provided by Docomo, a famous telecommunication company in Japan. This dataset divides Japan into numerous meshes and records the population in each mesh at different times (e.g., 0:00,

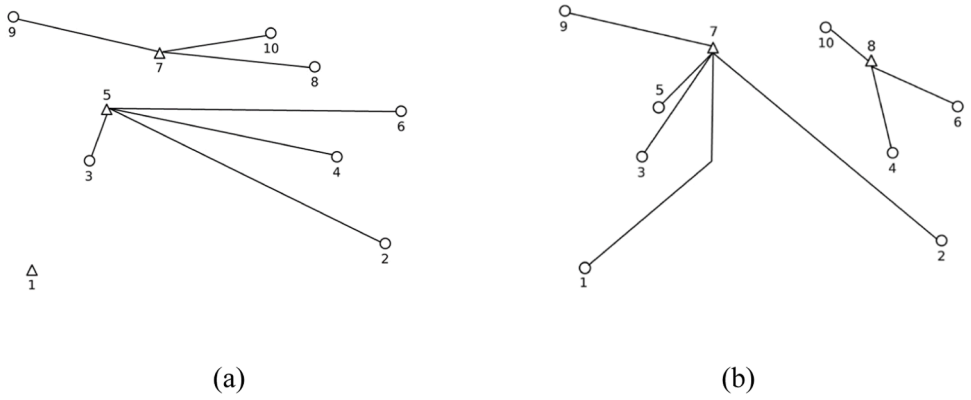


Fig. 11. Solutions for the deterministic (a) and stochastic (b) models on instance 5–10-T-L.

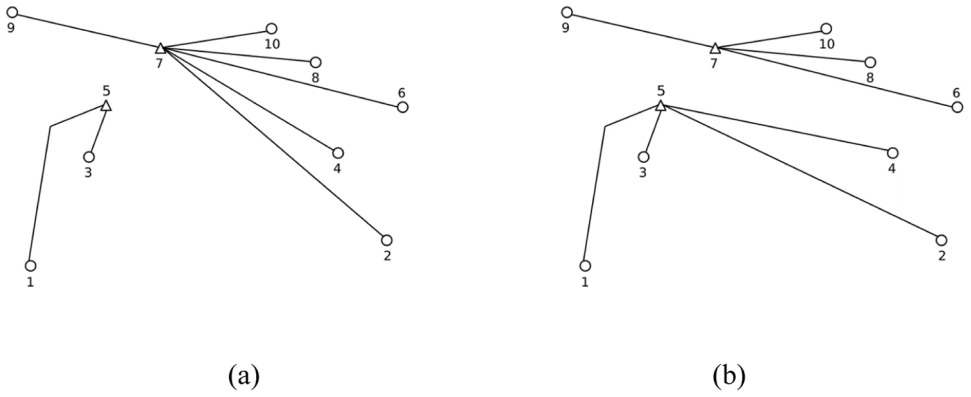


Fig. 12. Solutions for the deterministic (a) and stochastic (b) models on instance 5–10-L-T.

Table 7
Results of different TWFRs on instances with 20 clients.

		CHLRPTW			CHLRPTWSD			
Instance	TWFR	F_{det}	*Hub*	$T(s)$	F_{stc}	$Gap_{det}(\%)$	*Hub*	$T(s)$
5–20-L-L	0 %	392,210.97	4,5	79.19	364,769.89	7.52	4,5	427.62
	25 %	393,419.12	4,5	71.81	365,314.85	7.69	4,5	475.46
	50 %	410,596.76	4,5	64.92	365,536.02	12.33	4,5	443.39
	75 %	406,104.84	4,5	65.02	361,629.46	12.30	4,5	488.32
5–20-L-T	0 %	403,315.67	2,4	69.90	394,621.29	2.20	2,4	413.57
	25 %	403,111.55	3,4	75.60	395,845.98	2.05	2,4	481.63
	50 %	403,490.43	3,4	79.87	394,265.64	2.34	2,4	555.89
	75 %	402,874.31	3,4	70.88	394,033.91	2.24	2,4	475.59
5–20-T-L	0 %	476,987.34	2,4,5	140.35	414,097.09	15.19	2,4,5	352.88
	25 %	452,311.98	3,4,5	139.51	411,479.64	9.92	2,4,5	386.37
	50 %	462,022.79	2,4,5	140.35	411,036.98	12.40	2,4,5	441.48
	75 %	464,354.26	2,4,5	142.67	411,084.52	12.96	2,4,5	437.11
5–20-T-T	0 %	463,007.38	2,3,4	165.08	441,234.32	4.93	2,3,4	351.99
	25 %	484,498.42	2,3,4	160.47	440,026.62	10.11	2,3,4	391.71
	50 %	480,474.16	2,3,4	148.33	439,954.65	9.21	2,3,4	386.63
	75 %	528,578.53	2,3,4	149.71	439,816.37	20.18	2,3,4	377.73

1:00, and so on). The populations are further counted according to their district of residence, enabling the estimation of potential intra-city parcel flows between districts. OD flows are constructed as follows:

Step 1. Establish the relationship between the districts and meshes. Let *Mesh* and *District* denote the set of all meshes and districts, respectively; for each mesh $m \in Mesh$ and each district $i \in District$, the population pr^0_{im} from district i recorded in mesh m at 0:00. can be obtained from the dataset directly. Then, we identify the district $i \in District$ with the highest pr^0_{im} in mesh m as the district contains mesh m. For notation simplicity, we represent $\tau_{im} = 1$ if district i contains mesh m; otherwise $\tau_{im} = 0$;

Step 2. Obtain the population distribution at 0:00 for the districts. For each district $i \in District$ and each district $j \in District$, the population pp^0_{ij} recorded in district i at 0:00 that lives in district j is estimated via the meshes contained by district i, i.e., $pp^0_{ij} = \sum_{m \in Mesh} \tau_{im} pr^0_{jm}$.

Step 3. Obtain the population distribution at 14:00 for the districts. Similarly, the population pp^{14}_{ij} recorded in district i at 14:00 that

Table 8
Results of different TWFRs on instances with 25 clients.

		CHLRPTW			CHLRPTWSD			
Instance	TWFR	F_{det}	*Hub*	$T(s)$	F_{stc}	$Gap_{det}(\%)$	*Hub*	$T(s)$
5–25-L-L	0 %	422,562.59	2,5	152.29	381,458.69	10.78	2,5	972.63
	25 %	423,698.34	2,5	203.11	377,774.63	12.16	2,5	984.15
	50 %	420,326.59	2,5	187.37	373,072.37	12.67	2,5	1012.85
	75 %	484,653.43	2,5	196.16	372,277.95	30.19	2,5	1074.61
5–25-L-T	0 %	404,745.10	2,3	171.07	400,499.80	1.06	2,3	1011.20
	25 %	399,650.67	2,3	188.69	392,281.21	1.88	2,3	1087.24
	50 %	416,995.41	2,3	179.28	389,721.16	7.00	2,3	1086.67
	75 %	413,540.19	2,3	219.08	385,281.21	7.33	2,3	1114.32
5–25-T-L	0 %	464,226.55	2,4,5	342.31	449,220.05	3.34	2,4,5	893.70
	25 %	481,602.01	2,4,5	347.25	436,848.26	10.24	2,4,5	874.58
	50 %	481,973.58	2,4,5	332.28	435,716.19	10.62	2,4,5	834.58
	75 %	482,088.62	2,4,5	323.54	435,824.84	10.62	2,4,5	861.47
5–25-T-T	0 %	521,048.75	2,3,4	326.59	483,137.46	7.85	2,3,4	908.10
	25 %	544,423.33	2,3,4	320.40	478,544.11	13.77	2,3,4	897.13
	50 %	540,231.43	2,3,4	347.79	477,066.75	13.24	2,3,4	875.38
	75 %	540,580.46	2,3,4	294.77	475,541.24	13.68	2,3,4	901.57

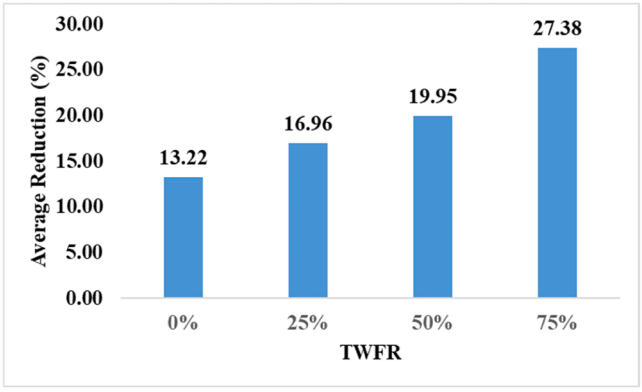


Fig. 13. Average reductions for instances with different percentages of time-window-free clients.

lives in district j can also be estimated using the same method as in **Step 2**.

Step 4. Generate population flows. For each pair of districts $i \in District$ and $j \in District$, if pp_{ij}^{0} is greater than pp_{ij}^{14}, then there is a population flow from district i to district j. Otherwise, there is a population flow from district j to district i.

Step 5. Build intra-city express ODs. We assume that the intra-city express flow is proportional to the population flow. For simplicity, a simple proportionality constant of 1 is applied, meaning each person contributes one unit of intra-city express demand. Thus, if $pp_{ij}^{0} > pp_{ij}^{14}$, then $d_{ij} = pp_{ij}^{0} - pp_{ij}^{14}$; else $d_{ji} = pp_{ij}^{14} - pp_{ij}^{0}$.

Three OD matrices are generated using the above method, based on the data of 11 districts in Kyoto, 14 in Yokohama, and 36 in Tokyo, respectively. The Yokohama case study is illustrated in Fig. 14. Five potential hubs (1, 3, 7, 11, 13) are selected from the 14 districts whose capacities and fixed costs are generated via the method proposed by Ebery et al. (2000). These potential hubs are selected according to their demands and geographical locations, i.e., we tend to choose districts

Table 9
Additional case study data.

Parameter	Definition	Value
α	unit backbone transportation cost	0.065
β	unit local tour cost	1
ω	unit overwork cost	1
f	vehicle fixed cost	4000
q	vehicle capacity	45

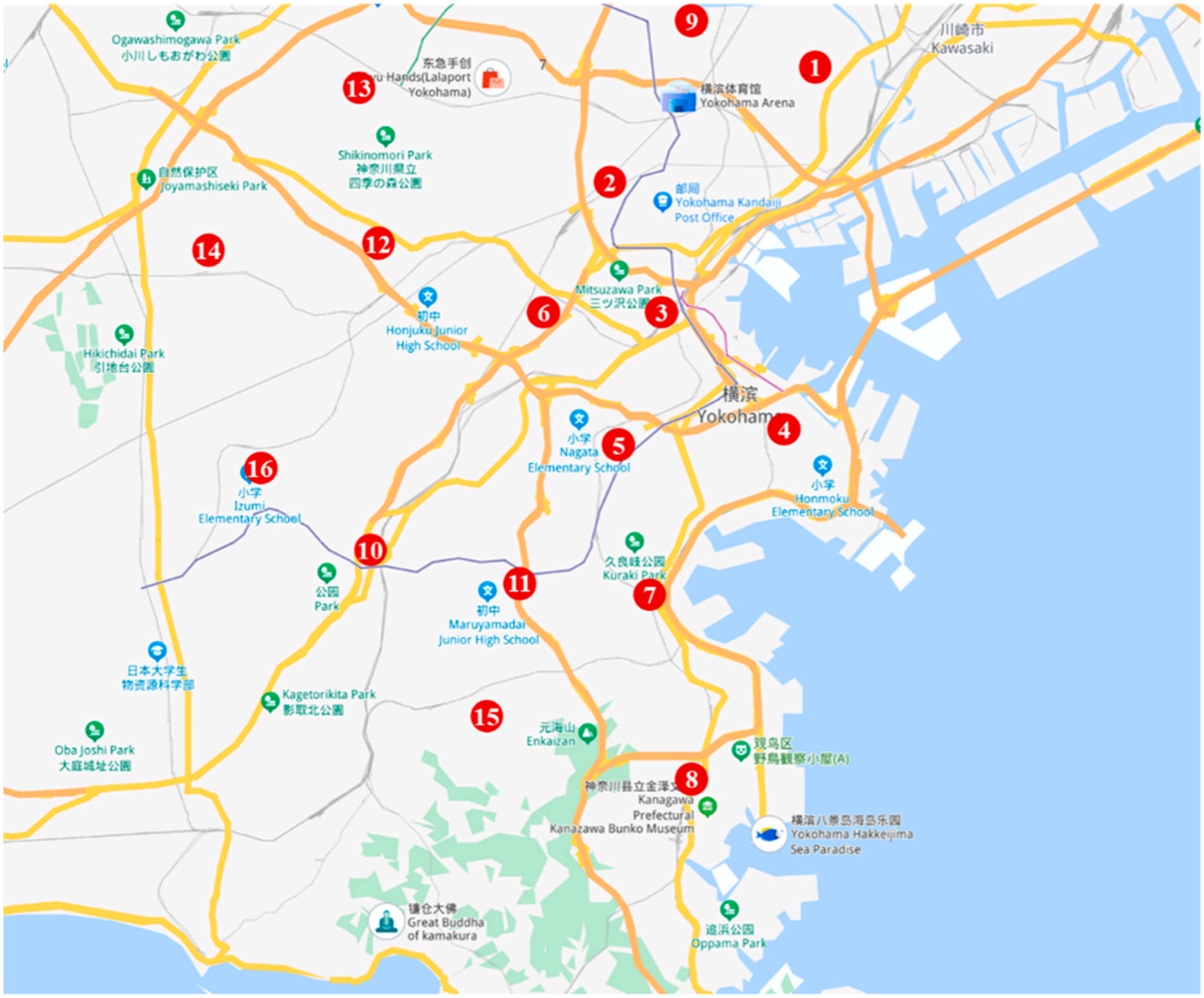


Fig. 14. 14 districts in the Yokohama case study.

Table 10
Case study results.

	CHLRPTW				CHLRPTWSD				
Instance	F_{det}	Hub(%)	CFR(%)	$T(s)$	F_{stc}	Gap_{det}(%)	Hub	CFR(%)	$T(s)$
Kyoto	124,751.36	1,8	30.51	15.74	105,214.26	15.66	1,8	6.59	56.44
Yokohama	172,963.21	3,11	21.73	59.51	158,462.96	8.38	3,10	10.21	86.32
Tokyo	275,184.62	7,14,26	19.22	956.33	241,242.32	12.33	7,14,26	9.66	6984.57

with more considerable demands and relatively far from the urban centre as potential hubs. These criteria can help reduce operation costs and the influence of intra-city express delivery on urban transportation. Moreover, time windows are defined using the approach provided in Section 5. Other parameters, such as unit backbone transportation cost and unit local tour cost, are detailed in Table 9.

Case study results are presented in Table 10 for the deterministic and stochastic models. It can be concluded from the table that incorporating stochastic demands can substantially reduce the operation cost (12.13 % on average and 15.66 % in the best case) and collection failure rate (CFR), which implies the possibility of collection failure in the evaluation sample (60.38 % on average and 78.40 % in the best case). These results indicate the potential benefits of employing the proposed model and algorithm in industrial applications, such as improving customer satisfaction.

7. Conclusions

We focused on designing an intra-city express system with several practical conditions taken into account, including hub and vehicle capacity, single allocation, time windows, and stochastic demands. The system was structured as a less-than-truckload hub-and-spoke network, and we proposed a CHLRPTWSD. The problem was formulated as a multi-stage recourse model, where long-term decisions (hub location and client-to-hub allocation) were determined first, followed by short-term decisions (vehicle routing) after the realisation of stochastic variables. An HSVNS algorithm, which embeds an ALNS algorithm within an SVNS framework, was proposed to solve the problem. Extensive numerical experiments indicated that the HSVNS algorithm could effectively solve the CHLRPTWSD for instances with up to 70 clients within reasonable computation times.

Additionally, incorporating stochastic factors could significantly reduce operation costs (8.11 % on average and 49.48 % in the best case), especially for instances with tight vehicle capacity and more time-window-free clients. Finally, case studies based on realistic data showcased the practical applicability of the proposed model and algorithm, highlighting its effectiveness in reducing the collection failure rate. Future research directions would include incorporating stochastic travel times, considering variable hub capacities, and solving the problem exactly.

CRediT authorship contribution statement

Yuehui Wu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Hui Fang:** Writing – review & editing, Methodology. **Ali Gul Qureshi:** Writing – review & editing, Methodology. **Tadashi Yamada:** Writing – review & editing, Supervision.

Declaration of interests

None.

Acknowledgement

This work was partially supported by the National Natural Science Foundation of China (Grant No 72301052), the Japan Society for the Promotion of Science (JSPS), Kakenhi (Grants-in-Aid for ScientificResearch - C) [20K04739], and the Fundamental Research Funds for the Central Universities (Grant No 3132024158).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2025.05.006.

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